

ANANYA SRIDHAR

AI/ML Engineer — Generative AI, LLMs & Agentic Systems

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PROFESSIONAL SUMMARY

Full-Stack AI/ML Engineer specializing in production-grade LLM architectures, multi-agent workflows (LangGraph), and deterministic RAG systems. Experienced in shipping end-to-end intelligent platforms across healthcare, legal, and fintech domains—combining fine-tuned transformers and reinforcement learning with high-throughput FastAPI backends and responsive React interfaces.

TECHNICAL SKILLS

- **AI & Deep Learning:** PyTorch, Hugging Face (Transformers, PEFT/LoRA, Accelerate), LangGraph, LangChain, ChromaDB, RAG, Reinforcement Learning (Q-learning, DRL), MCTS, Scikit-Learn
- **NLP & Speech:** Whisper, Vosk, Coqui TTS, Kokoro ONNX, Librosa, DistilBERT, BART, LIME, SHAP, Tokenization
- **Backend & Cloud:** Python, C++, SQL, FastAPI, Uvicorn, PostgreSQL, Supabase, Azure Cloud, Docker, WebSockets, REST APIs
- **Frontend & Web:** React.js, Vite, Next.js, Tailwind CSS, TypeScript, JavaScript, HTML5 Canvas

PROFESSIONAL EXPERIENCE

Manas Learning — AI/ML Engineer Intern (Full-Stack)

Mar 2026 – Present | Remote | Mumbai, India

- **LangGraph Diagnostic Pipeline:** Architected an agentic assessment workflow chaining Faster-Whisper, Vosk, and Kokoro TTS with phoneme-level acoustic verification (Librosa, Epitran) across 8 Indian languages.
- **Adaptive Reinforcement Learning:** Built a Q-learning difficulty controller dynamically adjusting practice game parameters in real time based on cumulative pronunciation accuracy and latency.
- **Full-Stack Architecture:** Engineered the FastAPI backend, PostgreSQL relational schemas, COPPA-compliant child privacy controls, and React + Vite clinical progress dashboards.

Nivesh AI — AI/ML Engineer Intern

Jun 2026 – Present | Mumbai, India

- **Enterprise RAG Pipeline:** Developed a zero-hallucination summarization engine on Azure processing advisor-client audio and portfolio documents for 10,000+ mutual fund distributors (MFDs).
- **Speech-to-Text Optimization:** Fine-tuned and evaluated multilingual STT models (Sarvam AI, Whisper) with spectral noise reduction (`noisereduce`), improving transcription accuracy on mixed Indian English/Hindi calls.
- **Structured Data Extraction:** Automated ingestion of complex PDF/DOCX financial portfolios into vector stores using `pdfplumber` and embedding models, accelerating advisor review turnaround by 80%.

FEATURED PROJECTS

NyayAI & AI Bail Reckoner — Legal-Reasoning LLM & Dense RAG

[GitHub: NyayAI](#) · [GitHub: Bail Project](#)

PyTorch, Hugging Face Transformers, PEFT/LoRA, ChromaDB, FastAPI, React

- Built a legal decision assistant predicting bail outcomes and retrieving precedent citations under the Indian Penal Code (IPC) and Bharatiya Nyaya Sanhita (BNS).
- Implemented ChromaDB dense vector retrieval to ground reasoning strictly in verified High Court and Supreme Court case laws, preventing generative hallucination.

Aegis — Multi-Agent Disaster Response Simulation

[GitHub: Aegis](#)

Python, NetworkX, NumPy, A Pathfinding, MCTS, Reinforcement Learning, FastAPI, React*

- Created a multi-agent simulation coordinating autonomous drone swarms, medical rescue teams, and ground vehicles under extreme environmental constraints.
- Formulated hazard-aware A* routing and Monte Carlo Tree Search (MCTS) for dynamic task allocation and casualty triage.

Explainable Misinformation Profiler — Real-Time NLP Framework

[GitHub: Misinformation-Profiler](#)

Hugging Face Transformers, DistilBERT, BART, LIME, SHAP, Next.js, Socket.io, FastAPI

- Built a streaming NLP pipeline ingesting live Reddit discourse via Socket.io for real-time stance and rumor classification.
- Integrated LIME and SHAP explainability modules to render token-level attribution heatmaps justifying model decisions.

Beshmai Gram — 3D Memory Palace & On-Device NLP Journal

[Live Demo: beshmai.vercel.app](#) · [GitHub](#)

Next.js, React Three Fiber, Gemini API, Supabase, Xenova Transformers (ONNX/Wasm)

- Built an interactive 3D spatial memory palace utilizing Gemini API for reflection synthesis and on-device Xenova Transformers for client-side, zero-latency sentiment classification in WebAssembly.

EDUCATION & CERTIFICATIONS

- **Bachelor of Technology (B.Tech)**, SASTRA University (Expected 2027)
- **SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems (*In-Person Program*)